When Better Offline Evaluation Still Cannot Choose the Better Policy

A practical lesson for reinforcement learning, contextual bandits, and AI deployment

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The Real Question Is Not the Estimate

Suppose an AI system already has a policy in production and a new candidate policy looks promising. The candidate might recommend different content, choose a different action in a contextual bandit, allocate resources differently, or change how an automated system reacts to a state. We would like to know whether the new policy is better before exposing users or operations to it.

That is exactly why off-policy evaluation (OPE) matters. It tries to evaluate a target policy from data generated by another behavior policy. But there is a subtle gap between estimation and deployment: a policy value can be estimated reasonably well while the difference between two close policies is still too uncertain to support a switch.

A better value estimate is not automatically a better policy decision.

The research behind this article studies that gap with a one-step contextual-bandit experiment. The empirical benchmark happens to use three highly liquid ETFs - QQQ, SPY, and IWM - because this setting lets us reconstruct all one-step action rewards after the next return is observed. The market data are the test bed, not the message. The broader question is about offline decision systems: when does logged evidence actually resolve which policy should be deployed?

A Truth-Auditable Offline Decision Benchmark

At each decision time, the policy chooses one of three actions: reduce exposure, keep it unchanged, or increase it by ten percentage points. In notation, the action set is

$$A = \{-1, 0, +1\}.$$

A Thompson-sampling behavior policy generates the partial-feedback log. Learning and OPE see only the reward of the sampled action, just as they would in a normal logged bandit problem. The other two one-step rewards are withheld.

The useful feature is that under the price-taking assumption, the next return does not depend on the small simulated exposure change. Once that return is observed, the experimenter can reconstruct the one-step reward for all three actions at the same pre-action state. This gives a withheld full-action benchmark for scoring the OPE estimates. It does not reconstruct a different multi-period path, so the claim is deliberately one-step and local.

This design makes the experiment unusually easy to audit: the estimator sees partial feedback, while the experimenter later knows the one-step answer well enough to measure both value error and whether the candidate-baseline decision was actually resolved.

1. More Recent Data Are Not Automatically Better Data

In nonstationary environments, a natural instinct is to discard old observations. Recent data feel more relevant. The problem is that a shorter window also throws away evidence.

The study holds policy construction fixed and varies only the historical evidence supplied to doubly robust OPE. Four pre-specified evidence windows are compared: all available pre-anchor history, the last 252 observations, the last 126, and the last 60.

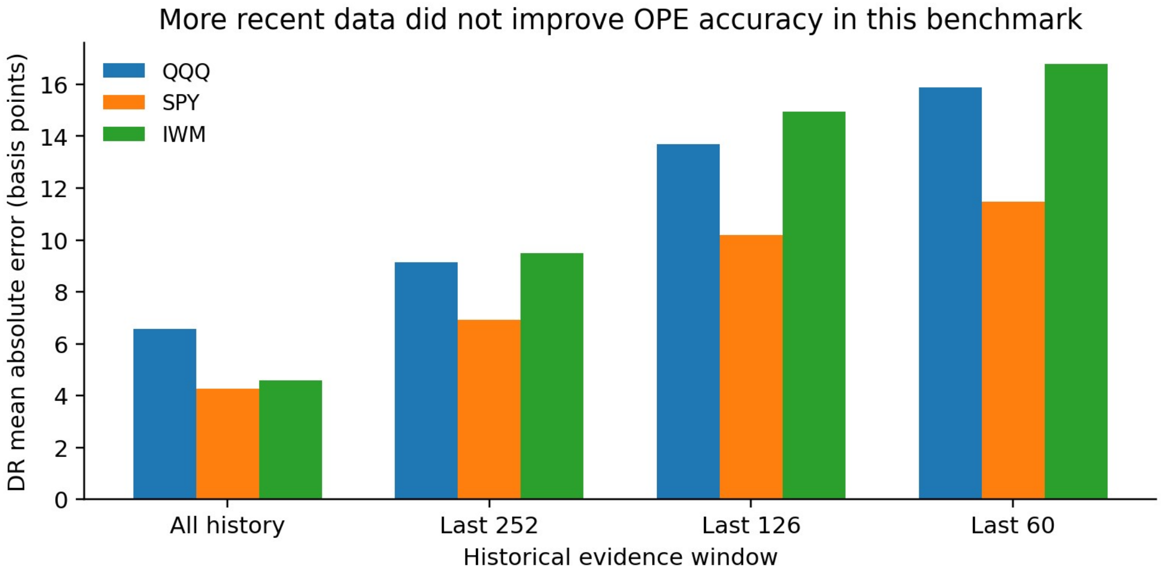

Figure 1. DR mean absolute error under four pre-specified historical evidence windows. In this benchmark, all available pre-anchor history gives the lowest error for QQQ, SPY, and IWM.

The ordering is consistent across all three data paths. QQQ rises from 6.58 basis points of DR MAE with all history to 15.89 with the last 60 observations. SPY rises from 4.26 to 11.48, and IWM from 4.60 to 16.77.

This does not prove that old data are always better, and it does not imply stationarity. It says something narrower: within these fixed windows and this one-step design, the information lost by truncation was more damaging than the gain from temporal proximity.

Freshness and reliability are different properties of evidence.

2. Better Action Coverage Is Not the Same as Better Evaluation

Recency is only one problem. A log can be recent and still be weak evidence for a target policy if the behavior policy rarely took the actions that the target now prefers.

For a target policy pi and behavior policy mu, the usual importance ratio is

rho_i(pi) = pi(a_i | x_i) / mu(a_i | x_i).

A common overlap diagnostic is effective sample size (ESS). The study tests action coverage separately from recency by taking the same recent contexts and re-logging actions from a behavior distribution shifted toward the baseline and candidate target policies.

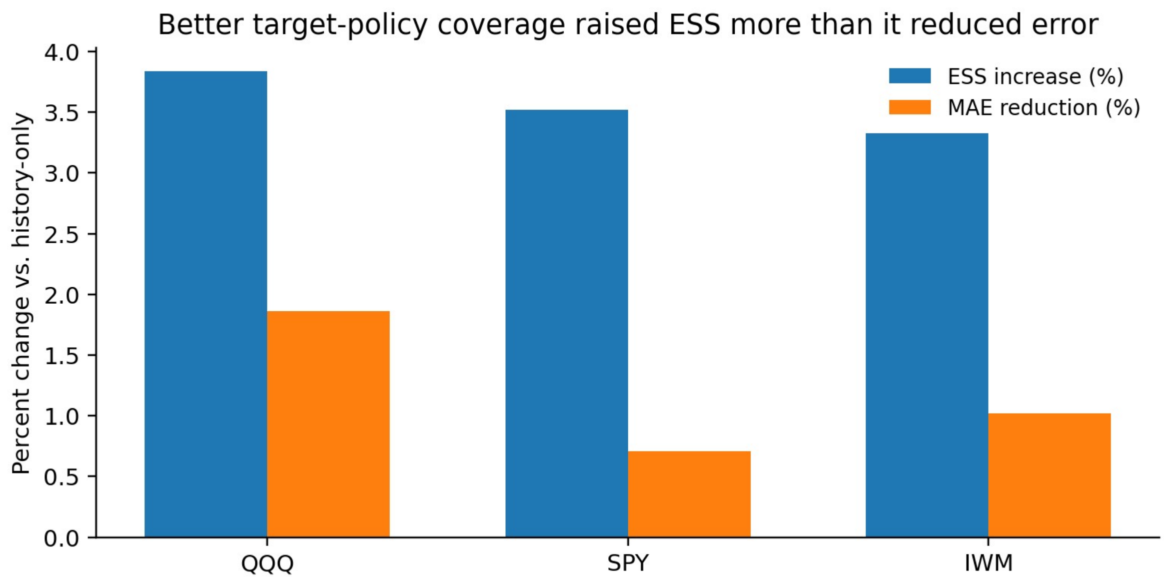

Figure 2. At the same recent contexts, shifting action coverage toward the target policies raises effective sample size more than it reduces OPE error.

At m = 80 recent observations, target-aware re-logging raises ESS by 3.84% for QQQ, 3.52% for SPY, and 3.33% for IWM. The corresponding DR MAE reductions are only 1.86%, 0.71%, and 1.02%.

The result is not that overlap is unimportant. Better overlap is real and useful. The point is that ESS measures concentration in target-behavior weights; it does not measure temporal transfer, and it does not directly tell us whether two close policies can be distinguished.

Better support can be real and still be insufficient for the decision we care about.

3. The Hardest Problem May Be Policy Resolution

The main result is not about choosing between a 60-day and 252-day history window. It is about the scale of the decision itself.

For each evaluation block, the study compares a baseline policy with a candidate policy and evaluates the paired difference

Delta V = V(candidate) - V(baseline).

The uncertainty is estimated for this paired contrast with a moving-block bootstrap. That matters because deployment depends on the sign and size of the difference, not on two marginal value estimates viewed separately.

Case	Median |gap|	Median SE	Gap/SE	Selection accuracy
QQQ	0.31 bp	3.42 bp	0.09	50.1%
SPY	0.26 bp	1.87 bp	0.14	50.2%
IWM	0.35 bp	1.86 bp	0.19	48.8%

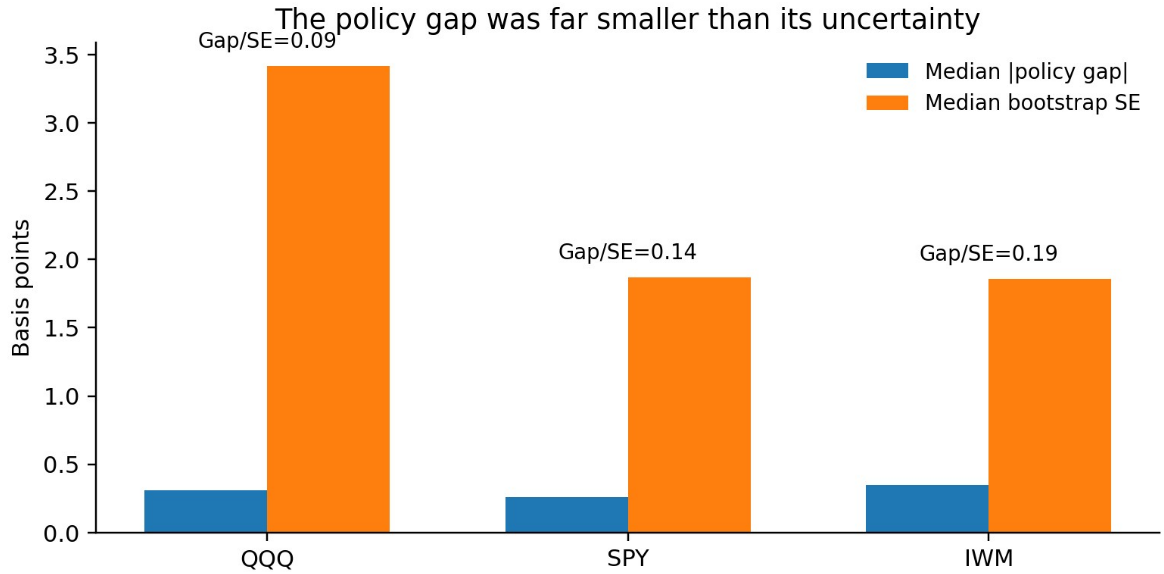

Figure 3. The candidate-baseline value gap is much smaller than its paired bootstrap standard error. Typical uncertainty is roughly five to eleven times the policy difference that must be resolved.

This is the decisive scale mismatch. The median policy gap is only 0.26-0.35 basis points, while the paired standard error is 1.86-3.42 basis points. Point-estimate selection stays near 50% across all three cases. A numerical ranking is always available, but the logged evidence does not contain enough resolution to make that ranking reliable.

The paper also applies an approximate one-sided 95% bootstrap lower-confidence rule. It admits the candidate in only about 2-3% of evaluation windows and otherwise retains the baseline. That low admission rate should be read as an evidence result: the log usually cannot justify a switch. It is not a certified 95% safety guarantee; the underlying paper explicitly does not claim a finite-sample coverage theorem for the full data-adaptive protocol.

Estimation accuracy and decision resolution are different statistical objects.

Why This Matters Beyond the Benchmark

The same distinction appears in many AI systems. A recommender may have two candidate policies whose estimated click-through values are close. A contextual bandit may have a new exploration policy that looks slightly better offline. A resource-allocation agent may improve an average reward estimate while the paired difference between old and new behavior remains dominated by uncertainty.

In each case, the deployment question is not "Can I estimate a value?" but "Can my evidence resolve the policy contrast I actually need to act on?" This is especially important when teams compare many small policy variants. Better MAE, a larger ESS, or a more recent dataset can all be useful diagnostics without changing the final decision.

This is also why abstention deserves a more prominent place in offline reinforcement learning and decision systems. If the data do not resolve the sign of the candidate-baseline gap, retaining the baseline is not a failure of the evaluator. It is a statement about what the evidence can support.

What the Estimator Benchmark Says - and What It Does Not Say

The study compares Direct Method, IPS, SNIPS, and doubly robust evaluation. No estimator dominates across all three data paths: SNIPS has the lowest MAE for QQQ and SPY, while DM has the lowest MAE for IWM. DR is not the lowest-MAE method in that benchmark.

DR is retained for the main analysis because it was pre-specified before the benchmark results were inspected. That prevents the paper from turning into post-hoc estimator shopping. More importantly, the scientific result does not depend on claiming that one estimator wins. The important mismatch is between the scale of the policy difference and the uncertainty of the evidence used to compare it.

A Practical Checklist for Offline Policy Evaluation

1. Separate recency from sample size. Do not assume that newer data are automatically more useful. Test whether the gain in temporal relevance compensates for the loss of evidence.

2. Check target-policy action support. Inspect importance ratios and ESS, but do not treat overlap as a universal score of evaluation quality.

3. Keep nuisance estimation time ordered. Predictions used inside OPE should be trained only on information available before the observation being evaluated.

4. Compare policies as a paired decision problem. Estimate the candidate-minus-baseline difference directly instead of judging two marginal values separately.

5. Measure resolution, not only error. Ask whether the policy gap is large relative to the uncertainty of that gap.

6. Allow abstention. If the evidence cannot resolve the policy contrast, "not enough evidence to switch" is a valid deployment decision.

What This Study Does Not Claim

The benchmark is deliberately narrow. The full-action reference is one-step and conditional on the logged pre-action position; it does not reconstruct the multi-period trajectory that would have followed from a different sequence of actions. The behavior logs are simulated rather than collected from a production system. The target policies are data-adaptive, and the three market paths are not independent draws from a general population of environments.

The evidence-window result should also not be read as "always use all history." Other environments, horizons, or structural changes could favor shorter windows. The paper is a controlled empirical demonstration, not a universal impossibility theorem.

Those limitations matter because they define the real contribution: a decision-first way to audit historical OPE. The experiment shows that conventional diagnostics can improve while the deployment comparison remains unresolved.

From Research Paper to Reproducible Practice

The full research paper contains the frozen protocol, target-policy construction, Thompson-sampling logger, time-ordered nuisance estimation, moving-block bootstrap, robustness checks, and reproducibility record. This article keeps only the pieces needed to understand the decision problem.

Historical logs -> recency audit -> coverage audit -> paired policy uncertainty -> deploy or abstain

Full research paper / arXiv: version forthcoming.

Code and reproducibility materials will be released through: https://github.com/SequentialDecisionResearch

Conclusion

Offline evaluation is not one-dimensional. More history can improve value accuracy. Better action coverage can improve effective sample size. Neither improvement guarantees that a close candidate-baseline choice has become resolvable.

The most useful question is therefore the last one: after all the estimation work, is the policy gap large enough relative to uncertainty to support a real decision? If the answer is no, a responsible system should be able to abstain.